

Response surface alignment measure

1 Set-up

AOD at the pair of gridboxes at longitude -171.5625 and latitude 51.875 and longitude -169.6875 and latitude 51.875 (shown in green and blue respectively in Figure 1) are considered, for July.

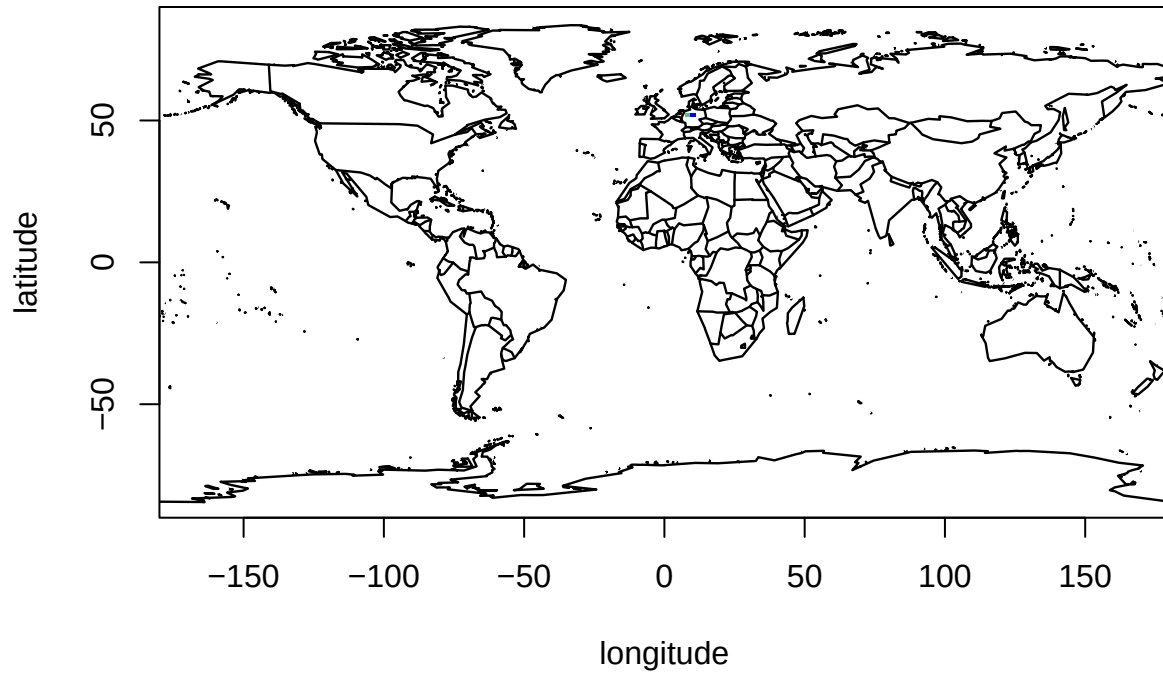


Figure 1: Selected gridboxes highlighted in green and blue.

2 Sensitivity analysis

Sensitivity analysis is not carried out.

3 Emulator fit and validation

To fit the emulator, we assume a linear mean function of the form

$$m(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p.$$

The Gauss (squared exponential) kernel is used for the covariance function, maximum posterior mode estimation for the parameters estimates (trend, range, SD and the nugget term), and the **RobustGaSP** package to fit the emulators.

Table 1: Estimated parameters

Ranges, trends, variance and nugget	Output 1	Output 2
1	5335	5404
2	4861	4833
3	4424	4436
4	5525	5550
5	5852	5834
6	4882	4902
7	7566	7635
8	5746	5643
9	4958	4940
10	4837	4798
11	5141	5137
12	5432	5376
13	4839	4828
14	4685	4691
15	5065	5109
16	5252	5247
17	5159	5151
18	4932	4921
19	5832	5731
20	4903	4961
21	5105	5123
22	5365	5360
23	4739	4755
24	5436	5426
25	5428	5400
26	5353	5396
27	4894	4850
28	5767	5727
29	4513	4550
30	5167	5189
31	4947	4950
32	5975	6007
33	6303	6292
34	4745	4718
35	5685	5725

Ranges, trends, variance and nugget	Output 1	Output 2
36	4896	4935
37	5475	5407
38	0	0
1	0	0
2	0	0
3	0	0
4	0	0
5	0	0
6	0	0
7	0	0
8	0	0
9	0	0
10	0	0
11	0	0
12	0	0
13	0	0
14	0	0
15	0	0
16	0	0
17	0	0
18	0	0
19	0	0
20	0	0
21	0	0
22	0	0
23	0	0
24	0	0
25	0	0
26	0	0
27	0	0
28	0	0
29	0	0
30	0	0
31	0	0
32	0	0
33	0	0
34	0	0
35	0	0
36	0	0
37	0	0
1	23	24
1	0	0

[1] 0.0001

For validation of the emulators fitted, as well as calculation of the alignment measure, 10000 new LHC points are generated and predictions made at these points. The mean width of the 95% CI's for these predictions are shown in Table 3, along with the mean width of the LOO CV prediction 95% CI's. Note that the range of output 1 is 0.6 and of output 2 is 0.6.

Table 2: Validation checks

Output	Prediction 95% CI mean width	LOO CV 95% CI mean width	% validated in LOO CV
1	0.23	0.23	95
2	0.24	0.24	96

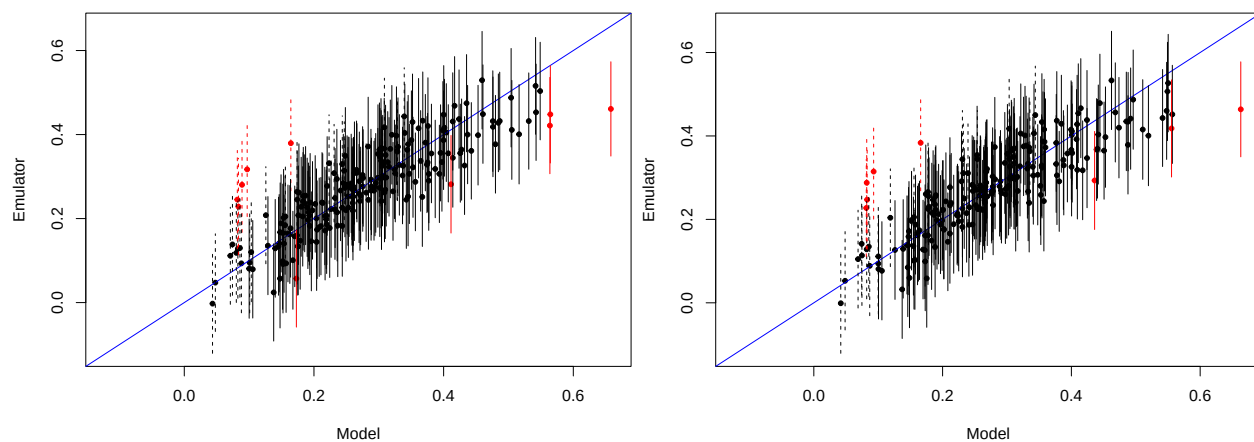


Figure 2: LOO CV validation plot for output 1 (left) and output 2 (right).

4 Check of by-hand calculations

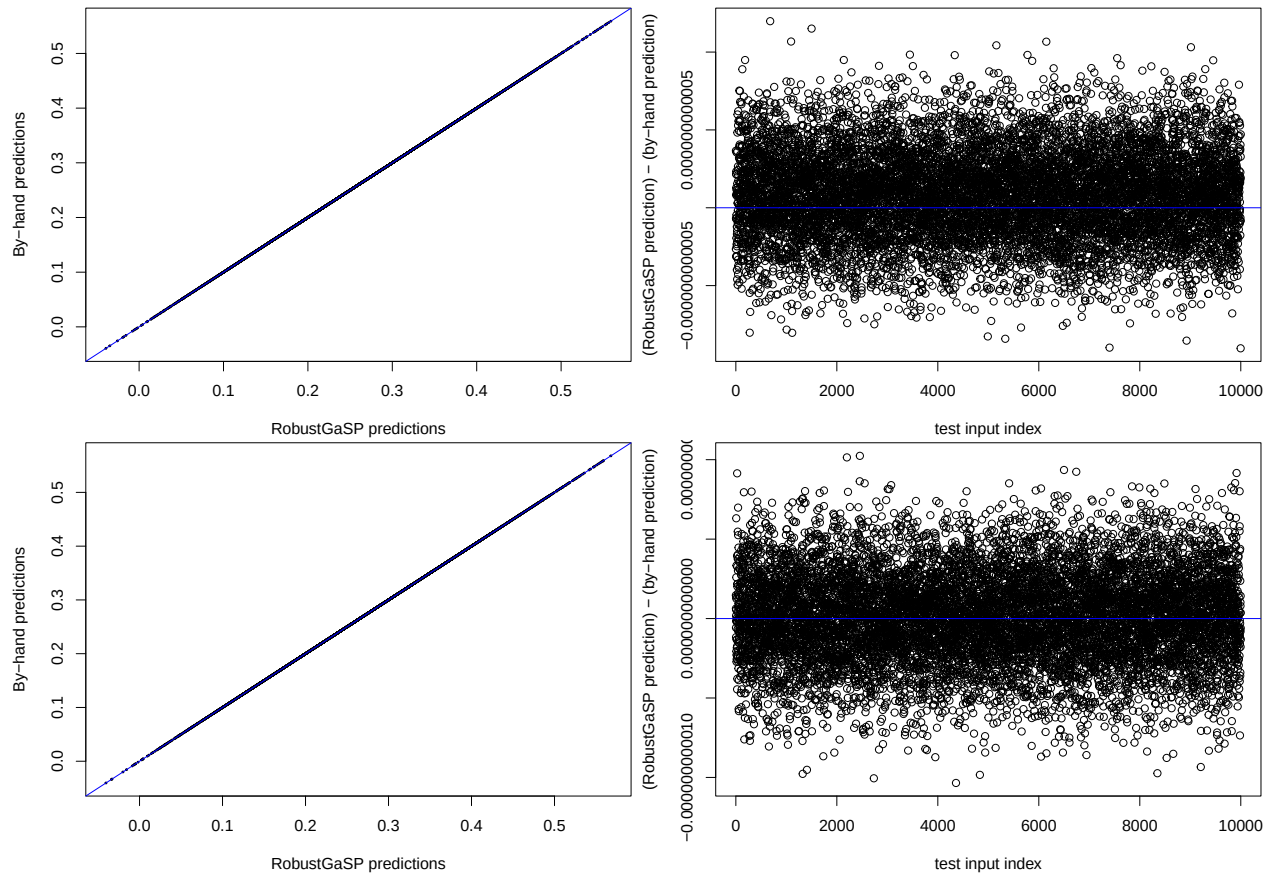


Figure 3: The predictions found by-hand match well with those made by the RobustGaSP package (left), with the latter minus the former found to be essentially zero (right). Top: Output 1. Bottom: Output 2.

Ahead of performing by-hand calculations of the partial derivatives at the testing points, by-hand calculations for the predictions made at these points are plotted against the predictions returned by the **RobustGaSP** package (left hand plots in Figure 3) and the difference between the two plotted against the test input index (right hand plots in Figure 3). Finally, for both we calculate

$$\frac{\sum_{i=1}^N \left| (\text{predictions from package})_i - (\text{predictions by-hand})_i \right|}{N},$$

where N is the number of test points to get 0 for output 1 and 0 for output 2.

5 Calculation of the alignment measure

Having calculated the emulated partial derivatives for use in the alignment measure, and normalising these, we calculate the AM, getting a value of 1.